Determinants

→ Determinants are always square (n×n)
→ It always has a scalar value.
→ Value of 1×1 determinant ⇒

$|a| = a$ i.e $|-3| = -3$ (with sign)

#Minors and Cofactors:

Eg: $D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

$M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$

* Cofactor, $C_{ij} = (-1)^{i+j} \cdot M_{ij}$

i.e $C_{11} = (-1)^2 \cdot M_{11} = M_{11}$

$C_{12} = (-1)^3 \cdot M_{12} = -M_{12}$

$C_{13} = (-1)^4 \cdot M_{13} = M_{13}$

$D = \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$

Expanding a 3×3 determinant ⇒

↔ We can expand it w.r.t any Row or column.

→ Value of det. can be obtained by taking summation of product of elements of a row/column & their corresponding cofactor

i.e $\Delta = \sum a_{ij} \cdot C_{ij}$

Expansion,

w.r.t Row-1 ⇒ $a_{11} \cdot C_{11} + a_{12} \cdot C_{12} + a_{13} \cdot C_{13}$

w.r.t Column-2 ⇒ $a_{12} \cdot C_{12} + a_{22} \cdot C_{22} + a_{32} \cdot C_{32}$

w.r.t Row 3 ⇒ $a_{31} \cdot C_{31} + a_{32} \cdot C_{32} + a_{33} \cdot C_{33}$

Cofactor Property ⇒

Sum of product of elements of any row/column and cofactors of some other row/colum is always zero.

i.e $a_{11} \cdot C_{21} + a_{12} \cdot C_{22} + a_{13} \cdot C_{23} = 0$

$a_{13} \cdot C_{11} + a_{23} \cdot C_{21} + a_{33} \cdot C_{31} = 0$

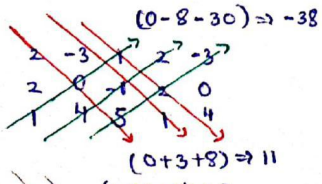
Shortcut for 3×3 Determinant ⇒

(i) Rewrite first two columns after the third column.

(ii) Now, first multiply down-wise diagonally, then multiply up-wise diagonally and substract.

Eg: $D = \begin{vmatrix} 2 & -3 & 1 \\ 2 & 0 & -1 \\ 1 & 4 & 5 \end{vmatrix} = 11 - (-38) \Rightarrow \boxed{49}$

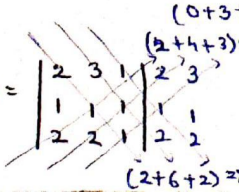
Sol^n $D =$



$(0-8-30) \Rightarrow -38$

$(0+3+8) \Rightarrow 11$

Eg: $D = \begin{vmatrix} 2 & 3 & 1 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{vmatrix}$



$(2+4+3) \Rightarrow 9$

$(2+6+2) \Rightarrow 10$

$D = 10 - 9 = +1$

#Properties of Determinants:

1) $|A^T| = |A|$

2) $|I| = 1$ always, for any order

3) If any two rows/columns of a determinant are interchanged, sign of det. changes.

$$\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = - \begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix}$$

4) If a determinants has two identical rows/columns, then $\Delta = 0$.

$$\begin{vmatrix} a & b & c \\ a & b & c \\ x & y & z \end{vmatrix} = 0 \quad \text{and} \quad \begin{vmatrix} a & a & x \\ b & b & y \\ c & c & z \end{vmatrix} = 0$$

5) Scalar multiplication takes place only with one row/column at a time.

$$k \begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix} = \begin{vmatrix} ka & x & p \\ kb & y & q \\ kc & z & r \end{vmatrix} = \begin{vmatrix} ka & kx & kp \\ b & y & q \\ c & z & r \end{vmatrix}$$

6) $|kA| = k^n |A|$ (n → order of determinant)

7) $|AB| = |A| \cdot |B|$

8) If $\det(A) = 0$ then A is called singular matrix.

9) Addition of Determinants ⇒

↳ It is possible only when any two rows/columns of the determinants to be added are identical.

i.e $\begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} p & q & r \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} x+p & y+q & z+r \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

10) Value of determinant remains same upon elementary row/column transformations.

i.e $R_1 \to R_1 \pm nR_2 \pm mR_3$
$C_1 \to C_1 \pm nC_2 \pm mC_3$

Coefficient of Row/column being transformed should be 1.

Que $D = \begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix}$

Sol^n $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$

$$D = \begin{vmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 3a & 7a+3b \end{vmatrix}$$

$$= a\left[a(7a+3b) - 3a(2a+b)\right]$$

$$= a\left[7a^2 + 3ab - 6a^2 - 3ab\right]$$

$$= a^3$$ Ans

Que Find solution of the equation

$$\begin{vmatrix} 1+\sin^2 x & \sin^2 x & \sin^4 x \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\sin 2x & 4\sin 2x & 1+4\sin 2x \end{vmatrix} = 0$$

Sol^n $R_1 \to R_1 + R_2$

$$\begin{vmatrix} 2 & 2 & 2 \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\sin 2x & 4\sin 2x & 1+4\sin 2x \end{vmatrix} = 0$$

$C_1 \to C_1 - C_2$

$$\begin{vmatrix} 0 & 2 & 2 \\ -1 & 1+\cos^2 x & \cos^2 x \\ 0 & 4\sin 2x & 1+4\sin 2x \end{vmatrix} = 0$$

$\Rightarrow -1\left[(1+\cos^2 x)(1+4\sin 2x) - \cos^2 x(4\sin 2x)\right] = 0$

$1 + 4\sin 2x + 4\cos^2 x \cdot \sin 2x = 4\cos^2 x \cdot \sin 2x$

$\Rightarrow 1\left[2(1+4\sin 2x) - 1(4\sin 2x)\right] = 0$

$2 + 4\sin 2x = 0$

$\Rightarrow \sin(2x) = -\frac{1}{2} \Rightarrow 2x = \pi + \frac{\pi}{6}, \; 2\pi - \frac{\pi}{6}$

$x = \frac{7\pi}{12} / \frac{11\pi}{12}$ Ans

Que Let $A = \begin{vmatrix} [x+1] & [x+2] & [x+3] \\ [x] & [x+3] & [x+3] \\ [x] & [x+2] & [x+4] \end{vmatrix}$. If $\det(A) = 192$ then $x \in$ ____.

Sol^n $A = \begin{vmatrix} [x]+1 & [x]+2 & [x]+3 \\ [x] & [x]+3 & [x]+3 \\ [x] & [x]+2 & [x]+4 \end{vmatrix} = 192$

$C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_2$

$A = \begin{vmatrix} [x]+1 & 1 & 1 \\ [x] & 3 & 0 \\ [x] & 2 & 2 \end{vmatrix} = 192$

$C_3 \to C_3 - C_2$

$A = \begin{vmatrix} [x]+1 & 1 & 0 \\ [x] & 3 & -3 \\ [x] & 2 & 0 \end{vmatrix} = 192$

$3\left[2[x] + 2 - [x]\right] = \frac{64}{192}$

$[x] + 2 = 64 \Rightarrow [x] = 62$

So, $x \in [62, 63)$ Ans

Que The number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval $-\frac{\pi}{4} \le x \le \frac{\pi}{4}$ is ____.

Sol^n $R_1 \to R_1 + R_2 + R_3$

$$(\sin x + 2\cos x) \begin{vmatrix} 1 & 1 & 1 \\ \cos x & \sin x & \cos x \\ \cos x & \sin x & \sin x \end{vmatrix} = 0$$

$C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$

$$(\sin x + 2\cos x)\begin{vmatrix} 1 & 0 & 0 \\ \cos x & \sin x - \cos x & 0 \\ \cos x & 0 & \sin x - \cos x \end{vmatrix} = 0$$

$$(\sin x + 2\cos x)(\sin x - \cos x)^2 = 0$$

$\sin x = -2\cos x$ | $\sin x = \cos x$

$\tan x = -1$ | $\tan x = 1$

In $-\frac{\pi}{4} \le x \le \pi/4$, tan can never be -2, thus No solution from here | $\Rightarrow \boxed{x = \pi/4}$

Thus, $\boxed{\text{No. of Roots} = 1}$ Ans

Special Determinants :

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$$

$$\begin{vmatrix} 1 & x & x^3 \\ 1 & y & y^3 \\ 1 & z & z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)$$

Circulant

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3+b^3+c^3-3abc)$$

Solving System of Linear Equations :

Cramer's Rule

- $D \neq 0$ (Unique Solution)
- $D = 0$
  - Atleast any one of D_1, D_2 or D_3 is $\neq 0$ ⇓ (No Solution)
  - $D_1 = D_2 = D_3 = 0$
    - If planes are parallel ⇓ (No solution)
    - If planes are not parallel ⇓ (∞ solution)

* Two planes are parallel when they satisfy

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \neq \frac{d_1}{d_2}$$

$a_1x + b_1y + c_1 = d_1$
$a_2x + b_2y + c_2 = d_2$
$a_3x + b_3y + c_3 = d_3$
} All constants (d) must be on RHS.

Then, $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

$D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$, $D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$, $D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$

Important Terms :→

1) Consistent ⇒ Solution exists (Unique or Infinite)
2) Inconsistent ⇒ Solution does not exist
3) Homogeneous Eqns ⇒ All constant terms are zero
4) Trivial Solution ⇒ All variables zero i.e. (0,0,0)

Homogeneous Linear Equations :→

$a_1x + b_1y + c_1 = 0$
$a_2x + b_2y + c_2 = 0$
$a_3x + b_3y + c_3 = 0$

Since, $d_1 = d_2 = d_3 = 0$
So, for homogeneous eq'ns $\boxed{D_1 = D_2 = D_3 = 0}$ always.

→ Thus, these system of eq'ns can have only 2 possibilities.

- $D = 0$
  - * Infinite soln
  - * Non-Trivial soln
  - * Consistent system
- $D \neq 0$
  - * Unique soln
  - * Trivial soln
  - * Consistent

Ques The system of eqn has no soln, if $K =$ ____.

$Kx + y + z = 1$
$x + Ky + z = K$
$x + y + Kz = K^2$

Soln $D = \begin{vmatrix} K & 1 & 1 \\ 1 & K & 1 \\ 1 & 1 & K \end{vmatrix} = (K+2)(K-1)^2 = 0$

$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ K & K & 1 \\ K^2 & 1 & K \end{vmatrix} = -(K-1)^2(K+1)$

For no soln, $D = 0$ but atleast one of $D_1, D_2, D_3 \neq 0$

For, $[K=1]$ $D=0$ and $D_1 = 0$ (x)
$[K=-2]$ $D=0$ and $D_1 \neq 0$ (✓)

Ans

Ques Find the values of K for which following system has unique soln & no soln.

$x - 2y = 1 \quad | \quad x - y + Kz = -2 \quad | \quad Ky + 4z = 6$

Soln $D = \begin{vmatrix} 1 & -2 & 0 \\ 1 & -1 & K \\ 0 & K & 4 \end{vmatrix} = (2-K)(2+K)$

$D_1 = \begin{vmatrix} 1 & -2 & 0 \\ -2 & -1 & K \\ 6 & K & 4 \end{vmatrix} = -(K+2)(K+10)$

$D_2 = \begin{vmatrix} 1 & 1 & 0 \\ 1 & -2 & K \\ 0 & 6 & 4 \end{vmatrix} = -6(K+2)$

$D_3 = \begin{vmatrix} 1 & -2 & 1 \\ 1 & -1 & -2 \\ 0 & K & 6 \end{vmatrix} = 3(K+2)$

For Unique Soln

$D \neq 0 \Rightarrow (2-K)(2+K) \neq 0$

i.e. $\boxed{K \in R - \{2, -2\}}$ Ans

For No Soln

$D = 0$ & atleast one of $D_1, D_2, D_3 \neq 0$

For $\boxed{K = 2}$, $D = 0$ and $D_2, D_3, D_3 \neq 0$

Ans

Characteristic Equation of a Sq. Matrix :

For any square matrix,

$$\boxed{|A - \lambda I| = 0}$$ (Characteristic eqn of A)

→ Values of λ are called 'Characteristic Roots'

Cayley - Hamilton Theorem ⇒

Any sq. matrix A, satisfies its characteristic equation, always.

Ques $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $A^{-1} = \frac{1}{6}(A^2 + cA + dI)$,

Then find (c+d).

Soln $|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -2 & 4-\lambda \end{vmatrix} = 0$

$0 = (1-\lambda)[(1-\lambda)(4-\lambda) + 2]$

$0 = (1-\lambda)[4 - 5\lambda + \lambda^2 + 2]$

$0 = 6 - 5\lambda + \lambda^2 - 6\lambda + 5\lambda^2 - \lambda^3$

$0 = -\lambda^3 + 6\lambda^2 - 11\lambda + 6$

$\Rightarrow \boxed{\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0}$

$\boxed{A^3 - 6A^2 + 11A - 6I = 0}$

We are given,

$A^{-1} = \frac{1}{6}(A^2 + cA + dI)$

$A \cdot A^{-1} = \frac{1}{6}(A^3 + cA^2 + dA)$

$6I = A^3 + cA^2 + dA$

$\Rightarrow A^3 + cA^2 + dA - 6I = 0$

On comparing

$c = -6$ & $d = 11$ $\Rightarrow \boxed{c+d=5}$ Ans